accumulated since the last impact off into space. So around 4 billion years ago, there was a fairly narrow window—perhaps only a hundred million years wide—in which our most distant ancestors came to be. Once conditions permitted, life arose fast. Somehow.

The first living things very likely were inept, far less capable than the most humble microbe alive today—perhaps just barely able to make crude copies of themselves. But natural selection, the key process first coherently described by Charles Darwin, is an instrument of such enormous power that from the most modest beginnings there can emerge all the richness and beauty of the biological world.

Those first living things were made of pieces, parts, building blocks which had to come into being on their own—that is, driven by the laws of physics and chemistry on a lifeless Earth. The building blocks of all terrestrial life are called organic molecules, molecules based on carbon. Of the stupendous number of possible organic molecules, very few are used at the heart of life. The two most important classes are the amino acids, the building blocks of proteins, and the nucleotide bases, the building blocks of the nucleic acids.

Just before the origin of life, where did these molecules come from? There are only two possibilities: from the outside or from the inside. We know that vastly more comets and asteroids were hitting the Earth than do so today, that these small worlds are rich storehouses of complex organic molecules, and that some of these molecules escaped being fried on impact. Here I'm describing homemade, not imported, goods: the organic molecules generated in the air and waters of the primitive Earth.

Unfortunately, we don't know very much about the composition of the early air, and organic molecules are far easier to make in some atmospheres than in others. There couldn't have been much oxygen, because oxygen is generated by green plants and there weren't any green plants yet. There was probably more hydrogen, because hydrogen is very abundant in the Universe and escapes from the upper atmosphere of the Earth into space better than any other atom (because it's so light). If we can imagine various possible